N

higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 4

25 FEBRUARY 2019

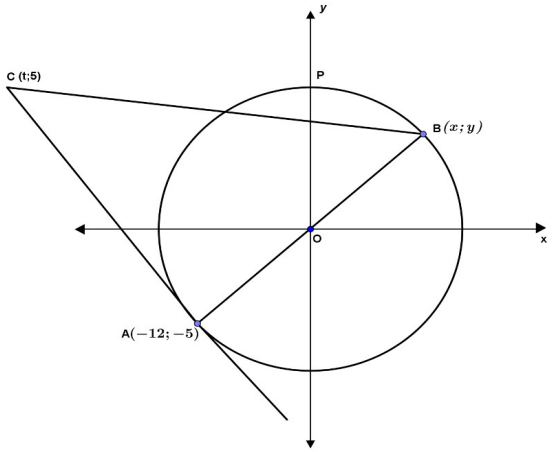
This marking guideline consists of 9 pages.

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Note to markers

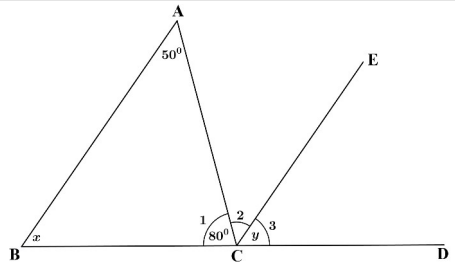
- Accept ALL mathematically valid solutions.
- Consistent accuracy will apply as a general rule
- If a candidate answers a question twice and does not delete either, mark the FIRST attempt.
- If a candidate answers a question, crosses it out and does not re-do it, mark the deleted attempt.

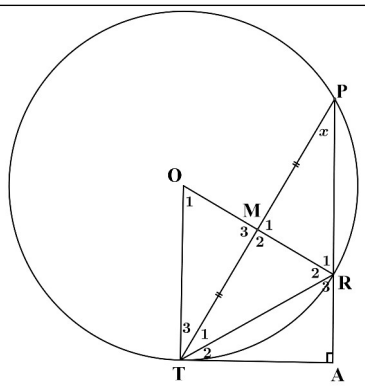
MARKING LEGEND	
SYMBOL	**EXPLANATION OF SYMBOL**
A	Accuracy
K	Knowing
M	Method
CA	Consistent Accuracy (follow-up)
B/D	Break down
S	Statement
R	Reason

	SOLUTION	EXPLANATION	
1.1			
1.1.1	$OA=\sqrt{(-12)^2+(-5)^2}=\sqrt{169}$ $OA=13$ ✓	1 for correct answer.	(1)
1.1.2	$x^2+y^2=169$ ✓	1 for correct answer	(1)
1.1.3	$m_{OA}=\frac{y}{x}=\frac{5}{12}$ ✓	1 for correct answer	(1)
1.1.4	$m_{AC}=-\frac{12}{5}$ ✓ and point $P(-12;-5)$ $y-(-5)=-\frac{12}{5}(x-(-12))$ $y=-\frac{12}{5}$ ✓ $x-\frac{169}{5}$ ✓	1 for m_{AC} 1 for correct m 1 for correct c	(3)
1.1.5	Substitute $C(t;5)$ into $y=-\frac{12}{5}x+7$ $5=-\frac{12}{5}t-\frac{169}{5}$ ✓ $t=-\frac{97}{6}$ ✓	1 for substitution (CA) 1 for answer (CA)	(2)
1.1.6	$O(0;0)$ is the midpoint of B	1 for correct x	(2)

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	$\frac{x+(-12)}{2}=0 \Rightarrow x=12$ ✓ $\frac{y+(-5)}{2}=0 \Rightarrow y=5$ ✓	1 for correct y	

1.2			
1.2.1	$50^0+x+80^0=180^0$ $\therefore x=50^0$	1 for correct answer	(1)
1.2.1	$A\hat{C}D=50^0+x$ ✓$^{\frac{1}{2}}=100^0$ … ext angle of Δ $2y=100^0$ ✓$^{\frac{1}{2}}$ … angle bisected $\therefore y=50^0$ ✓	½ for text angle (CA) ½ for $2y$ (CA) 1 for answer (CA)	(2)
1.2.3	Yes. ✓ CE // BA $x=y=50^0$ ✓$^{\frac{1}{2}}$ and x and y are corresponding angles. ✓$^{\frac{1}{2}}$	1 for answer (CA) ½ for each part of reason	(2)

1.3			
1.3.1	$\hat{A}=90°$ ✓$^{\frac{1}{2}}$ … *given* $\hat{M}_2=90°$ ✓$^{\frac{1}{2}}$ ... *line ¿ centre ¿ midpt of chord* $\therefore \hat{M}_2+\hat{A}=180°$ ✓$^{\frac{1}{2}}$ $\therefore MTAR$ *is a cyclic* ... opp angles supplementary ✓$^{\frac{1}{2}}$	½ for correct answer ½ for correct answer ½ for sum ½ for reason	(2)
1.3.2	Δ *RPM* $\equiv \Delta$ *RTM* ✓ … side, angle, side ✓ $\therefore PR=RT$ ✓	1 for Δ's, 1 for reason 1 for conclusion	(3)
1.3.3	$T_1=P=x$ … PR = TR $T_2=P=x$ … tan chord theorem TR bisects $P\hat{T}A$	½ for S, ½ for R ½ for S, ½ for R	(2)
1.3.4	$O_1=2x$ ✓$^{\frac{1}{2}}$ … angle at centre thm ✓$^{\frac{1}{2}}$	½ for S, ½ for R	(1)

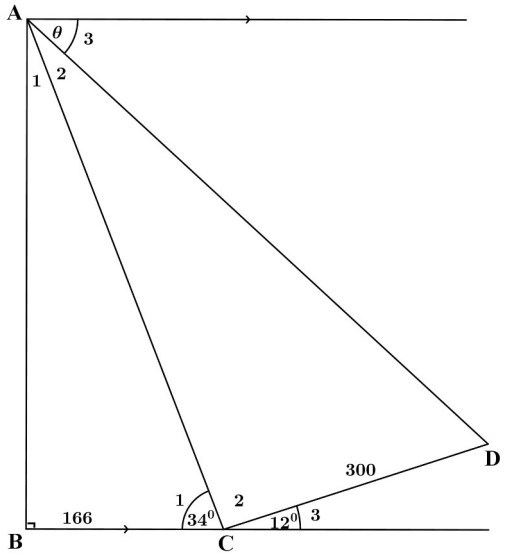
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1.3.5	$P\hat{T}A = T_1 + T_2 = 2x$ ✓½ … from 1.3.3 Also, $P\hat{T}A = 90^0 - x$ ✓½… Δ PTA is right angled at T $\therefore 2x = 90^0 - x$ ✓½ $3x = 90^0$ $x = 30^0$ ✓½	½ for S ½ for S ½ for equating ½ for correct answer	(2)
			[25]

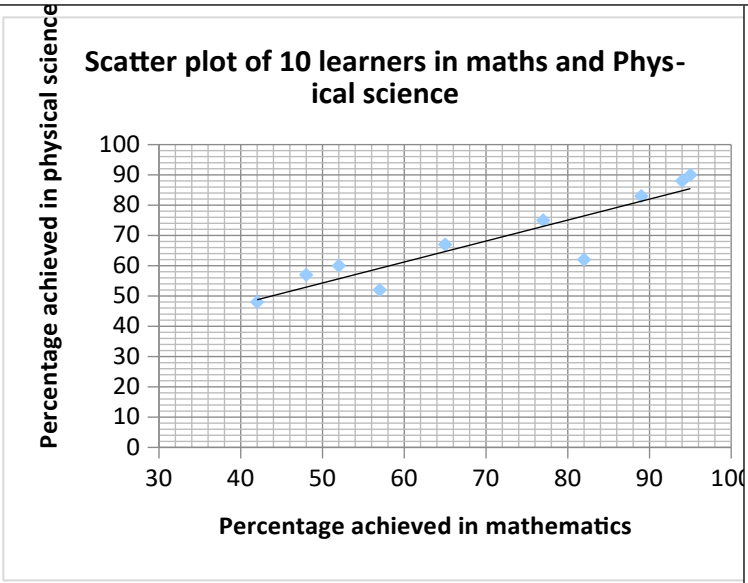
2.1.1	cos (A + B) = cos A cos B – sin A sin B Let B = A: cos (A + A) = cos A cos A – sin A sin A✓ ∴ cos 2A = cos² A – sin² A But sin² A = 1 – cos² A✓ ∴ cos 2A = cos² A – (1 – cos² A) = cos² A – 1 + cos² A = 2 cos² A – 1✓	1 for substituting A for B 1 for using identity 1 for conclusion	(3)
2.1.2	From QUESTION 2.1.1: cos 2A = cos² A – sin² A But cos² A = 1 – sin² A✓ ∴ cos 2A = 1 – sin² A – sin² A = 1 – 2 sin² A✓	1 for using identity 1 for conclusion	(2)
2.2	$\cos 2x - 7\cos x - 3 = 0$ $2\cos^2 x - 1 - 7\cos x - 3 = 0$ $2\cos^2 x - 7\cos x - 4 = 0$ $(2\cos x + 1)(\cos x - 4) = 0$ $\therefore \cos x = -\frac{1}{2}$ ❑½ $\vee \cos x = 4$ ❑½ $(not\ \ valid)$ Reference angle: $x = 60°$ $\therefore x = 180° - 60° = 120°$ $\therefore x = 180° + 60° = 240°$ or	1 for subst for $\cos 2x$ 1 for factorising ½ each for $\cos x$ values 1 for answer (CA) 1 for answer (CA)	(5)
2.3	$(\sin 75° - \cos 75°)(\sin 75° + \cos 75°)$ $\sin^2 75° - \cos^2 75°$ ❑½ $-(\cos^2 75° - \sin^2 75°)$ ❑½ $= -[\cos 2(75°)]$ ❑½ $= -[\cos 150°]$ ❑½ $= -\left(\frac{-\sqrt{3}}{2}\right)$ ❑½ ¿ $\frac{\sqrt{3}}{2}$ ❑½	½ for multiplying ½ for rearranging ½ for identity ½ for $\frac{-\sqrt{3}}{2}$ (CA) ½ for answer (CA)	(3)

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2.4	$\sqrt{3}\ \tan 2\theta = 1$ $\tan 2\theta = \frac{1}{\sqrt{3}}$ ❑$^{\frac{1}{2}}$ $2\theta = 30^0$ ✓; 210^0 $\theta = 15^0$ ❑$^{\frac{1}{2}}$; 105^0 ✓ in the required interval	½ for simplifying 1 for 1st angle ½ first correct answers 1 for second correct answer	(3)

2.5			
2.5.1	$\widehat{C}_2 = 180° - (34° + 12°)$ $= 134°$	1 for correct answer	(1)
2.5.2	$\cos 34° = \frac{BC}{AC}$ AC $= \frac{BC}{\cos 34°}$ AC $= \frac{166}{\cos 34°}$ AC = 200m	1 for ratio 1 for correct answer	(2)
2.5.3	$AD^2 = AC^2 + CD^2 - 2AC \times CD \cos \widehat{C}_2$ $AD^2 = 200^2 + 300^2 - 2(200)(300)\cos 134°$ =213 359 m AD = 462 m	1 for formula 1 for correct subst (CA) 1 for answer (CA)	(3)

2.5.4	$\frac{\sin C\hat{A}D}{CD}=\frac{\sin 134°}{AD}\ \frac{\sin C\hat{A}D}{300}=\frac{\sin 134°}{462}$ $\sin C\hat{A}D=\frac{300\ \sin 134°}{462}\ C\hat{A}D\ =\ 27{,}8°$	1 for sine formula 1 for correct subst (CA) 1 for answer (CA)	(3)
			[25]

3.1		4 for 10 correct points 3 for 7 to 9 correct points 2 for 4 to 6 correct points 1 for 2 to 3 correct points 0 for 1 correct point	(4)
3.2	On graph	1 for line through most points	(1)
3.3	$\bar{x}=\frac{\sum x}{n}=\frac{52+82+94+95+65+77+42+89+48+57}{10}=$ $=70{,}1$ $s^2=\frac{\sum(x-\bar{x})^2}{n}$ $s^2=\frac{(52-70{,}1)^2+(82-70{,}1)^2}{10}+¿$ $\frac{(94-70{,}1)^2+(95-70{,}1)^2+(65-70{,}1)^2}{10}+¿$ $\frac{(77-70{,}1)^2+(42-70{,}1)^2+(89-70{,}1)^2+¿(48-70}{10}$ $=\ 354{,}09$	1 for correct mean	(3)

	$Standard\ \ Deviation(\sigma)=\sqrt{354,09}$ $=18,81$	1 for correct s^2 1 for correct answer	

3.4

Maths x	Physical Science y	xy	X^2	Y^2	
52	60	3120	2704	3600	½ mark as shown on table
82	62	5084	6724	3844	
94	88	8272	8836	7744	
95	90	8550	9025	8100	
65	67	4355	4225	4489	
77	75	5775	5929	5625	
42	48	2016	1764	2304	
89	83	7387	7921	6889	
48	57	2736	2304	3249	
57	52	2964	3249	2704	
$\sum x=701$	$\sum y=682$	$\sum xy=50259$ ✓	$\sum x^2=52681$ ✓½	$\sum y^2=$¿ 48548 ✓	
$\overline{x}=70,1$ ✓½	$\overline{y}=68,2$ ✓½				

$$b=\frac{n\sum xy-\sum x\sum y}{n\sum x^2-\left(\sum x\right)^2}$$

$$=\frac{10(50259)-(701)(682)}{10(52681)-(701)^2}$$

$$=\frac{502590-478082}{526810-491401}$$

$b=0,692$

1 for correct answer

$a=\overline{y}-b\overline{x}$

$=68,2-(0,692)70,01$

$a=19,68$ ✓½

½ for answer (CA)

Least squares regression line:

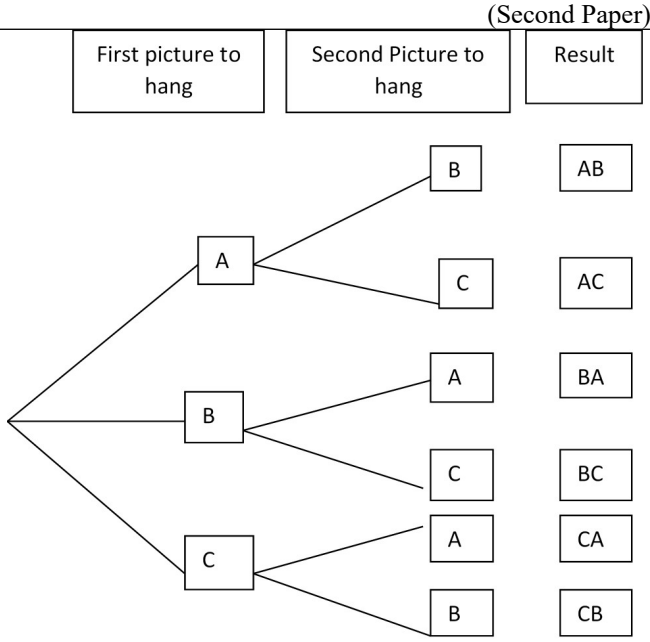
$y=19,68+0,692x$

1 for answer (CA)

(5)

3.5	$y = 19{,}68 + 0{,}692\,x$ $y = 19{,}68 + 0{,}692(60)$ $= 61{,}2$ The candidate would have scored 61,2%	1 for substitution (CA) 1 for answer (CA)	(2)
			[15]

4.1	INDEPENDENT EVENTS Independent events are events where the outcome of one event does not affect the outcome of the other events.	1 for comparing outcome 1 for does not effect, influence, etc	(2)
4.2.1	<table><tr><th>Gender</th><th>Drakensberg National Park</th><th>Cape Town Table Mountain</th><th>Pilanesberg Nature Reserve</th><th>Total</th></tr><tr><td>Female</td><td>18</td><td>16</td><td>**11** ❑½</td><td>45</td></tr><tr><td>Male</td><td>**16** ❑½</td><td>**25** ❑½</td><td>14</td><td>55</td></tr><tr><td>Total</td><td>**34** ❑½</td><td>41</td><td>**25** ❑½</td><td>**100** ❑½</td></tr></table>	½ for each correct entry	(3)
4.2.2	$P(F \text{ and } D) = \frac{18}{100} = 0{,}18$ $P(F)\times P(D) = \frac{45}{100} \times \frac{34}{100} = 0{,}153$ $P(F \text{ and } D) \neq P(F)\times P(D)$ Hence F and D are not independent ✓	1 for answer (CA) 1 for product (CA) 1 for conclusion (CA)	(3)
4.3.1		1 for first level outcome 1 for second level outcome ½ for each result as recorded for the branch. (the result need not be written if labels are correctly written on the branch)	(5)

			
4.3.1	P(AC or CA) ✓ $=\frac{2}{6}$ ✓$=\frac{1}{3}$	1 for two possibilities 1 for answer (CA)	(2) **[15]**

5.1.1	**BANK A:** $A=P(1+i)^{n}\square^{\frac{1}{2}}$ $A=50\ 000(1+0,08)^{4}\square^{\frac{1}{2}}$ $=R68024,45$ **BANK B_** $A=P(1+\in)\square^{\frac{1}{2}}$ $A=50\ 000(1+0,12\times4)\ \square^{\frac{1}{2}}$ $=R74000$ Bank B will be the better option.	½ correct formula ½ for correct substitution 1 for correct answer ½ correct formula ½ for correct substitution 1 for correct answer 1 for conclusion (CA)	(5)
5.1.2	**BANK A:** $A=P(1+i)^{n}$ $A=50\ 000(1+0,08)^{15}$ $=R158608,46$ **BANK B_** $A=P(1+\in)$ $A=50\ 000(1+0,12\times15)$ $=R140000$ Bank A will be a better option.	1 for correct answer 1 for correct answer 1 for conclusion (CA)	(3)
5.2	Total Income: **R265000,00 ✓** Pension: $1300\times12=15600\ \square^{\frac{1}{2}}$ UIF: $148.72\times12=1784,64\ \square^{\frac{1}{2}}$ Total non-taxable income: **R17384,64 ✓** Taxable income: **R247615,36✓** Tax: 34178+26% of (247615,36-189880) ✓ Normal Tax: **R49189.19 ✓** Less primary rebate (13635) **R35554,19 ✓** Med credits $(2\times303+2\times204)$ ✓ Total Med credits **R1014,00 ✓** Tax payable: **R34540,19✓** Tax paid to SARS (2900×12) **R34800,00✓** Amount **due to James**: **R259,81 ✓**	1 for total income ½ for total pension ½ for total UIF 1 for total non-taxable amt. 1 for taxable income 1 for tax calculation 1 for normal tax 1 for rebate deduction 1 med credit calculation 1 for total med credits 1 for tax payable 1 for amt. paid to SARS 1 for answer	(12)
			[20]
Total Marks: 100			

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